

Machine Learning Technique for Phishing Website Detection

Nurul Amira Binti Mohd Zin
Faculty of Computing
Universiti Malaysia Pahang Al-
Sultan Abdullah,
26600, Pekan, Pahang, Malaysia
nurulamira2617@gmail.com

Ferda Ernawan
Faculty of Computing
Universiti Malaysia Pahang Al-
Sultan Abdullah,
26600, Pekan, Pahang, Malaysia
ferda@ump.edu.my

*Mohd Faizal Ab Razak
Faculty of Computing
Universiti Malaysia Pahang Al-
Sultan Abdullah,
26600, Pekan, Pahang, Malaysia
faizalrazak@ump.edu.my

Nor Saradatul Akmar Zulkifli
Faculty of Computing
Universiti Malaysia Pahang Al-
Sultan Abdullah,
26600, Pekan, Pahang, Malaysia
saradatulakmar@ump.edu.my

Ahmad Firdaus
Faculty of Computing
Universiti Malaysia Pahang Al-
Sultan Abdullah,
26600, Pekan, Pahang, Malaysia
firdausza@ump.edu.my

Abstract— The Internet has emerged as an indispensable tool in both our personal and professional life in our modern day. As a direct consequence of this, the number of customers who make their purchases over the Internet is quickly increasing. Internet users may be vulnerable to a wide variety of web threats because of this fact. These threats may result in monetary loss, fraudulent use of credit cards, loss of personal data, potential damage to a brand's reputation, and customer mistrust in e-commerce and online banking. Phishing is a sort of cyber threat that may be defined as the practice of imitating a genuine website for the purpose of stealing sensitive information such as usernames, passwords, and credit card numbers. This research focuses on strategies for detecting phishing attacks. This study apply a machine learning approach to detect a phishing attack. As a result, this study able to detect phishing with accuracy 94%.

Keywords—*phishing attack; phishing; website detection; malware; machine learning*

I. INTRODUCTION

Phishing attacks are prevalent cyber threats that exploit any communication channel to trick individuals into divulging sensitive information. Attackers manipulate victims in fake scenarios to obtain data for personal harm or corporate damage. The attacker's objective and the data type determine the communication mode choice.[1] In addition, it involves ransom demands and threats of account cancellation. Another deceptive technique, email spoofing, targets customers, leading them to disclose sensitive information like credit card numbers and passwords. Phishing primarily aims to obtain critical data such as bank login credentials and credit card details. These fraudulent activities erode trust in online transactions, damaging the reputation of Internet organizations. Despite data encryption measures, computer systems remain vulnerable to attacks.[2] Awareness and vigilance are key to avoiding phishing attacks. Developing the habit of carefully browsing the Internet and verifying the trustworthiness of links can prevent harm. Browser extensions and technologies can alert users to fraudulent websites that aim to steal credentials. Implementing network systems that lock down everything except designated sites enhances

security, albeit at the expense of user convenience.[1] In this work, machine learning was used to identify phishing.

Phishing website detection techniques used are heuristic-based methods that collect information from websites to identify authenticity. Unlike blocklists, heuristics can detect phishing sites in real time during their construction. Successful heuristic methods rely on discriminatory criteria for differentiating website types. Using HTML or URL signatures, the heuristic approach identifies phishing websites. Ongoing studies are exploring the effectiveness of this method.[3]

In evaluating machine learning and data mining algorithms for predicting phishing sites, Logistic Regression (LR), Bayesian Additive Regression Trees (BART), Classification and Regression Trees (CART), Random Forests (RF), and Neural Networks (NN) are compared. Experiments with a dataset of 1171 phishing emails and 1718 real emails utilized 43 functions for training and testing classifiers. Results show RF with the lowest error rate of 7.72%, followed by CART (08.13%), LR (08.58%), BART (09.69%), Support Vector Machines (SVM) (09.90%), and NN (10.73%). However, according to the findings, no single classifier emerges as the best for predicting phishing sites.[4] The research evaluates and compares the effectiveness of machine learning-based detection methods (MLBDMs), including Bagging, AdaBoost, SVM, CART, NN, RF, LR, NB, and BART. The dataset comprises 1,500 phishing websites and 1,500 trustworthy websites. Evaluation factors conducted by CANTINA encompass a total of eight criteria.[4]

II. PHISHING WEBSITE ATTACK TRENDS

The Anti-Phishing Working Group (APWG) documented 316,747 attacks in the month of December 2021, making it the month with the greatest monthly total in the organization's history of reporting. Phishing scams have been more common since the beginning of the year 2020. In the fourth quarter of 2018, the financial industry was the industry that was the most often targeted by phishing, accounting for 23.2% of all attacks. The amount of cyberattacks directed against suppliers of SaaS and webmail has remained quite high. The percentage

of attacks that were phishing scams aimed at cryptocurrency targets, such as bitcoin exchanges and wallet providers, rose to 6.5%. The number of companies discovered to be ransomware victims jumped by 36% from the third quarter to the fourth quarter. 51.8% of emails identified as phishing attacks by business users were efforts to steal credentials, 38.6% of emails were response-based attacks (such as BEC, 419, and gift card scams), and 9.6% of emails were attempted to distribute malware.[5]

In addition, the APWG documented 316,747 attacks in the month of December 2021. This is the greatest monthly total in the history of the APWG's reporting. Phishing scams have been more common since the beginning of the year 2020. In the fourth quarter of 2018, the financial industry was the industry that was the most often targeted by phishing, accounting for 23.2% of all attacks. The amount of cyberattacks directed against suppliers of SaaS and webmail has remained quite high. The percentage of attacks that were phishing scams aimed at cryptocurrency targets, such as bitcoin exchanges and wallet providers, rose to 6.5%. The number of companies that were discovered to be victims of ransomware jumped by 36% from the third quarter to the fourth quarter. 51.8% of emails reported by business users were credential theft phishing attacks, 38.6% were response-based attacks (such as Business email compromise (BEC), 419, and gift card scams), and 9.6% were.[5]

III. WEBSITE VULNERABILITIES

This section discusses the most prevalent website vulnerabilities that result in phishing attacks: Cross-Site Scripting (XSS) XSS vulnerabilities occur when attackers inject malicious scripts into web pages viewed by users, potentially leading to the execution of unauthorized code in the victim's browser. This can allow attackers to redirect users to phishing websites or steal their login credentials. A comprehensive guide to understanding and mitigating XSS vulnerabilities can be found in the book "The Web Application Hacker's Handbook" by D. Stuttard and M. Pinto[6].

Next, Cross-Site Request Forgery (CSRF) vulnerabilities enable attackers to trick users into unintentionally performing actions on a targeted website without their knowledge or consent. This can include submitting forms or making transactions on phishing sites. The paper "Robust Defenses for Cross-Site Request Forgery" by A. Barth et al. provides insights into effective countermeasures against CSRF attacks.[7]

Moreover, SQL injection vulnerabilities arise when attackers can manipulate user-supplied input to execute malicious SQL queries against a website's database. Exploiting such vulnerabilities can lead to unauthorized access to sensitive user information, which can then be exploited in phishing attacks. [8] Then, session hijacking vulnerabilities occur when attackers gain unauthorized access to a user's session ID, enabling them to impersonate the user and perform malicious actions, including redirecting them to phishing websites. The paper "Session Hijacking and Its Countermeasures" by M. Naveed et al. [9] discusses session hijacking techniques and effective countermeasures.

IV. METHODOLOGY

This research approach is being used in this study since it can be reverted to prior stages with low loss to apply fresh research advancements. Not only that, but this process enables adjustments to any step to be made to tackle problems that arise during the current stage. Finally, this research approach allows researchers to readily adjust to the study topic's demands.[10]

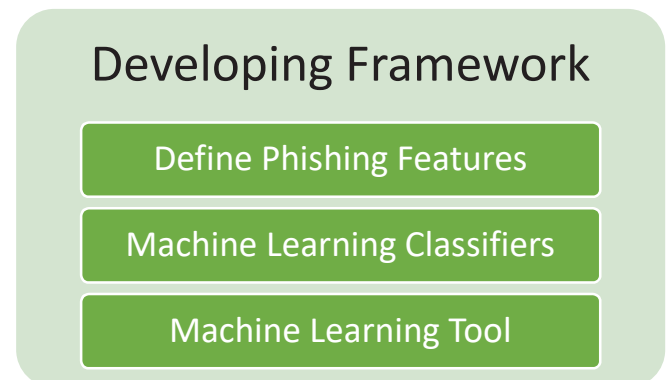


Fig. 1. Development of Phishing Website Detection Framework

A. Define Phishing Features

The primary emphasis will be on traits based on URLs. When determining whether to phish a website, the very first thing that should be looked at is the URL. Certain distinguishing traits may be found in phishing domain URLs. The URL is analysed in order to get characteristics that are associated with these points. The following URL-based features are going to be studied as part of this project:

- i. Address Bar-based Functionality
- ii. Abnormal Base Characteristics
- iii. HTML and JavaScript-based Functionality
- iv. Domain-specific Functions

B. Dataset Description

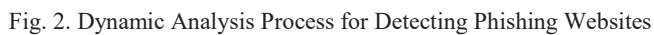
The dataset gathering is the first crucial step in implementing accurate phishing detection. It provides insights into both phishing and legitimate activities. The dataset consists of 48 characteristics extracted from 5000 authentic websites and 5000 fraudulent web pages. Unlike regular expression-based parsing, a browser automation framework is used for more reliable feature extraction. The categorical values "Legitimate" and "Suspicious" have been converted to numerical values, with "Legitimate" represented as "1" and "Suspicious" as "0." The dataset is analyzed using the Correlation Attribute Evaluation technique to determine the importance of each characteristic. This technique ranks the characteristics and provides their respective rank numbers. Several characteristics have achieved the highest ranking due to their frequent utilization in the detection process.[11][12]

C. Machine Learning Classifiers and Tools

Machine learning is a type of AI that acquires knowledge without explicit programming, predicting future outcomes and improving current ones with new evidence [31][32]. It searches for patterns in data sets to make predictions, known

Google Colab is used for data set training because of its adaptability and cloud capabilities. It is useful in the context of machine learning using Python. Optimizing performance by distributing GPU assets from Google servers to otherwise constrained hardware on the programmer end is critical to the memory-hogging machine learning algorithm. Google Storage provides a cloud drive infrastructure for storing this data set, which is then loaded and trained using the Colab online notebook. The trained model is then put into the Pi and validated using the collected data.[14][15][16]

Collecting data, identifying key factors, testing the model, and finally comparing the findings are the four components that make up the design model. Each component will get a cursory examination in the next subtopic.



The dynamic analysis technique involves four steps of empirical assessment: Fig. 2 illustrated the dynamic analysis process for detecting phishing websites. The first step in detecting phishing websites is by collecting phishing website datasets. Researchers frequently utilize various phishing website datasets in their studies and experiments. Some commonly used datasets include PhishTank, which is a collaborative platform for reporting and verifying phishing URLs[17]. OpenPhish provides a publicly available dataset containing a comprehensive collection of phishing URLs[18]. The Anti-Phishing Working Group (APWG) maintains a repository of phishing URLs reported by individuals and organizations, offering researchers a valuable

The dynamic analysis technique involves four steps of empirical assessment: Fig. 2 illustrated the dynamic analysis process for detecting phishing websites. The first step in detecting phishing websites is by collecting phishing website datasets. Recent studies have proved that the greater the number of datasets used in the experiment, the more accurate the result.[22], [23] Researchers frequently use phishing datasets such as Kaggle[24], Git[25]



FEATURES

Model	Features
NUMDOTS	24397
PATHLEVEL	5847
NUMDASH	32954
NUMQUERY ...	18167
NUMNUME ...	1389
DYAININP ...	4578
PATHLENGTH	2769
NUMSENSIT ...	58013
FREQUENT ...	222
IFRAMEORF ...	4282
PCTEXTRES ...	187880
PCTEXTNUL...	355196
NUMDOTS	85911
PATHLEVEL	1092
NUMDASH	8422
NUMQUERY ...	2153
NUMNUME ...	1286
DYAININP ...	3388
PATHLENGTH	9645
NUMSENSIT ...	6327
FREQUENT ...	3870
IFRAMEORF ...	6075
PCTEXTRES ...	701519

The following step is defining phishing features by extracting the most frequent data used from the dataset in Fig. 3 using correlation attribute evaluation which involves

determining the value of a characteristic by analyzing the degree to which the attribute relates to the class. It not only offers a ranking of the characteristics from best to worst but also shows the rank number for each quality[12]. According to Fig. 4, some of the characteristics have the greatest ranking since they are used the most often throughout the detection process.

Then, the third step is testing the chosen dataset which the experiment will be assessed when each component has been included into the whole. In order to address the problem statement and assess whether the limitation of current journals is avoided, testing and evaluation are carried out. The primary objective of this examination is to demonstrate the most effective detection model that has been suggested in order to validate the validity of the outcomes and assertions made in this investigation. In addition, the testing and evaluation process enables the research experiment to identify flaws and restrictions, which in turn makes it possible for further adjustments to achieve the desired conclusion.

The final step is to analyze the results using machine learning techniques such as random forest[26], J48[27], Naïve Bayes[28], KNN[29], and Logistic regression[30]. These methods extract insights and make predictions from the data. Random forest constructs multiple decision trees to generate accurate results. J48 is a decision tree classifier that identifies significant features and provides interpretable rules. Naïve Bayes is a probabilistic algorithm efficient for large datasets. KNN classifies instances based on proximity to known instances. Logistic regression models relationships between variables for binary or multi-class classification. These techniques uncover patterns, provide valuable insights, and facilitate informed decision-making.

VI. DISCUSSION

The outcomes present the results using three different machine learning classifiers: random forest, J48, Naïve Bayes, KNN, and Logistic. In addition, this investigation into the various measures made use of the metrics of accuracy, precision, and recall using Python. Table I shows the findings obtained from 25 phishing website features of the testing set that used five chosen classifiers.

Table I. The Related Study Classifiers Analysis

Classifiers	Accuracy	Precision	Recall	FPR	TPR
Random Forest	94.10%	0.978	0.904	0.021	0.904
J48	92.10%	0.917	0.926	0.084	0.926
Naïve Bayes	83.00%	0.921	0.771	0.071	0.771
KNN	92.21%	0.923	0.921	0.079	0.921
Logistic	89.50%	0.895	0.895	0.105	0.895

VII. CONCLUSION

In conclusion, this study plays a vital role in the overall investigation by contrasting prior solutions with the current Machine Learning approach for phishing website detection. The chapter highlights the methods employed in developing the proposed malware detection technique and emphasizes the

need for further improvement in existing strategies. The proposed approach presented in Chapter 3 aims to assist internet users in recognizing phishing websites effectively. Additionally, this chapter provides a comprehensive overview of the methodology employed throughout the research, including the necessary tools and instruments. The subsequent chapter will cover the installation, testing, and evaluation processes to ensure the success of the phishing website detection methodology.

Furthermore, the findings from Chapter 4 reveal that the Random Forest algorithm outperformed other algorithms in terms of accuracy, precision, True Positive Rate (TPR), and Receiver Operating Characteristic (ROC) values, with a remarkable accuracy rate exceeding 94%. J48 and KNN algorithms also exhibited consistent performance above 90%, while Naïve Bayes and Logistics showed slightly lower performance in certain tests. Based on these observations, it can be concluded that the Random Forest algorithm is the most suitable for detecting phishing attacks. The study emphasizes the significance of addressing security issues such as phishing in the context of the internet's transformative impact on human life. By focusing on machine learning-based phishing website detection, the research optimizes feature datasets, employs machine learning classifiers, and achieves high accuracy through the random forest classifier. The study also identifies areas for improvement, including feature selection, false alarm rate reduction, and exploration of dynamic analysis approaches. Future work should prioritize relevant feature selection and further enhance the detection methodology while considering dynamic analysis techniques.

ACKNOWLEDGEMENT

This work was supported by Universiti Malaysia Pahang Al-Sultan Abdullah, under the internal grant RDU210321.

REFERENCES

- [1] S. Hossain, D. Sarma, and R. J. Chakma, "Machine Learning-Based Phishing Attack Detection," 2020. [Online]. Available: www.ijacsa.thesai.org
- [2] M. N. Alam, D. Sarma, F. F. Lima, I. Saha, R. E. Ulfath, and S. Hossain, "Phishing attacks detection using machine learning approach," in *Proceedings of the 3rd International Conference on Smart Systems and Inventive Technology, ICSSIT 2020*, Institute of Electrical and Electronics Engineers Inc., Aug. 2020, pp. 1173–1179. doi: 10.1109/ICSSIT48917.2020.9214225.
- [3] N. Chou, R. Ledesma, Y. Teraguchi, and J. C. Mitchell, "Client-side defense against web-based identity theft." [Online]. Available: www.ebaymode.com
- [4] D. Miyamoto, H. Hazeyama, and Y. Kadobayashi, "An Evaluation of Machine Learning-based Methods for Detection of Phishing Sites."
- [5] "Phishing E-mail Reports and Phishing Site Trends 4 Brand-Domain Pairs Measurement 5 Brands & Legitimate Entities Hijacked by E-mail Phishing Attacks 6 Use of Domain Names for Phishing 7-9 Phishing and Identity Theft in Brazil 10-11 Most Targeted Industry Sectors 12 APWG Phishing Trends Report Contributors 13 4 th PHISHING ACTIVITY TRENDS REPORT," 2022. [Online]. Available: <http://www.apwg.org>.
- [6] Dafydd Stuttard; Marcus Pinto, "Dafydd Stuttard, Marcus Pinto - The web application hacker's handbook_ finding and exploiting security flaws- Wiley (2011)".
- [7] P. F. Syverson, Somesh. Jha, Xiaolan. Zhang, and A. Association for Computing Machinery. Special Interest Group on Security, CCS'08 : *proceedings of the 15th ACM Conference on Computer and Communications Security : Alexandria, Virginia, USA, October 27-31, 2008*. Association for Computing Machinery, 2008.

- [8] W. G. J. Halfond, J. Viegas, and A. Orso, "A Classification of SQL Injection Attacks and Countermeasures," 2006.
- [9] L. and M. A. Vishnoi, "International Journal of Computer Science & Information Security," *International Journal of Computer Science & Information Security*, vol. 15, pp. 1–425, 2013, [Online]. Available: <https://sites.google.com/site/ijcsis/>
- [10] "Research Methodology Methods and Techniques (PDFDrive)".
- [11] Choon Lin Tan, "Phishing Dataset for Machine Learning: Feature Evaluation," *Mendeley Data*, Mar. 24, 2018.
- [12] Betha Nurina Sari, "CorrelationAttributeEval," *ResearchGate*, Apr. 25, 2017.
- [13] F. Vanhoenshoven, G. Napoles, R. Falcon, K. Vanhoof, and M. Koppen, "Detecting malicious URLs using machine learning techniques," in *2016 IEEE Symposium Series on Computational Intelligence, SSCI 2016*, Institute of Electrical and Electronics Engineers Inc., Feb. 2017. doi: 10.1109/SSCI.2016.7850079.
- [14] Institute of Electrical and Electronics Engineers, IEEE Communications Society;, Denshi Jōhō Tsūshin Gakkai (Japan). Tsūshin Sosaiei, and Han'guk T'ongsin Hakhoe, *ICUFN 2019 : the 11th International Conference on Ubiquitous and Future Networks : July 2 (Tue.)-July 5 (Fri.) 2019, Zagreb, Croatia*.
- [15] M. Kuroki, "Using Python and Google Colab to teach undergraduate microeconomic theory," *International Review of Economics Education*, vol. 38, p. 100225, Nov. 2021, doi: 10.1016/J.IREE.2021.100225.
- [16] Prabanjan Raja, "What is Google Colab?," *Scaler Topics*, Feb. 11, 2022.
- [17] PhishTank, "Phishing Websites Database," May 2023.
- [18] OpenPhish, "OpenPhish Dataset," May 2023.
- [19] "Anti-Phishing Working Group (APWG)," May 2023.
- [20] Kaggle, "Phishing Websites Dataset," May 2023.
- [21] GitHub, "Phishing Datasets," May 2023.
- [22] N. Mallios, E. Papageorgiou, M. Samarinas, and K. Skriapas, "Comparison of machine learning techniques using the WEKA environment for prostate cancer therapy plan," in *Proceedings of the 2011 20th IEEE International Workshops on Enabling Technologies: Infrastructure for Collaborative Enterprises, WETICE 2011*, 2011, pp. 151–155. doi: 10.1109/WETICE.2011.28.
- [23] B. Espinoza, J. Simba, W. Fuertes, E. Benavides, R. Andrade, and T. Toulkeridis, "Phishing attack detection: A solution based on the typical machine learning modeling cycle," in *Proceedings - 6th Annual Conference on Computational Science and Computational Intelligence, CSCI 2019*, Institute of Electrical and Electronics Engineers Inc., Dec. 2019, pp. 202–207. doi: 10.1109/CSCI49370.2019.00041.
- [24] Kaggle, "<https://www.kaggle.com>," Jun. 2023.
- [25] G. Vrbančič, I. Jr. Fister, and V. Podgorelec, "Datasets for Phishing Websites Detection," 2020.
- [26] L. Breiman, "Random Forests," 2001.
- [27] by J. Ross Quinlan, M. Kaufmann Publishers, and S. L. Salzberg, "Programs for Machine Learning," 1994.
- [28] I. Rish and I. Rish, "An Empirical Study of the Naïve Bayes Classifier Predicting conversion to psychosis in clinical high risk patients using resting-state functional MRI features View project Clinical Machine Learning based on Cardiorespiratory models and simulation View project An empirical study of the naive Bayes classifier." [Online]. Available: <https://www.researchgate.net/publication/22884526> 3
- [29] N. S. Altman, "An Introduction to Kernel and Nearest-Neighbor Nonparametric Regression," 1992.
- [30] S. L. R. X. S. David W. Hosmer Jr., "Applied Logistic Regression," 2013, doi: 10.1002/9781118548387.
- [31] A. Zabidi, M. I. M. Jaya, W. I. S. W. Din, H. A. Hassan, and I. M. Yassin, "Non-Linear Autoregressive with Exogenous Input (Narx) Chiller Plant Prediction Model," *Proc. - 2021 Int. Conf. Softw. Eng. Comput. Syst. 4th Int. Conf. Comput. Sci. Inf. Manag. ICSECS-ICOCSIM 2021*, vol. 1, pp. 388–393, 2021.
- [32] N. S. Zaini, D. Stiawan, A. F. Mohd Faizal Ab Razak, S. K. Wan Isni Sofiah Wan Din, and T. Sutikno, "Phishing detection system using machine learning classifiers," *Indones. J. Electr. Eng. Comput. Sci.*, vol. 17, no. 3, pp. 1165–1171, 2020.